

Css3-Day-3 Hands on -Gurunath Rageni

Problem 1

Assessment Goal: Check ability to apply styling and understand selectors and box model.

Hands-on Tasks:

1. Create an external CSS file and link it to HTML
2. Apply different text styles to headings and paragraphs
3. Add background color and borders to a section
4. Demonstrate margin and padding on a div
5. Style the form created on Day 2

Code:

```
<!DOCTYPE html>

<html>

<head>

  <title>Styled Registration Form</title>

  <link rel="stylesheet" href="style.css">

</head>

<body>


  <header>

    <h1>Student Registration</h1>

    <p>Apply styling using external CSS</p>

  </header>


  <section class="form-section">


    <div class="box-model-demo">

      This div demonstrates margin and padding.

    </div>


    <form>

      <label>Full Name:</label>
```

```
<input type="text" placeholder="Enter your name">
```

```
<label>Email:</label>
```

```
<input type="email" placeholder="Enter your email">
```

```
<label>Course:</label>
```

```
<select>
```

```
  <option>HTML</option>
```

```
  <option>CSS</option>
```

```
  <option>JavaScript</option>
```

```
</select>
```

```
<br><br>
```

```
<input type="submit" value="Register">
```

```
</form>
```

```
</section>
```

```
</body>
```

```
</html>
```

```
/* Universal Selector */
```

```
* {
```

```
  font-family: Arial, sans-serif;
```

```
}
```

```
/* Element Selector */
```

```
h1 {
```

```
  color: darkblue;
```

```
    text-align: center;
}
```

```
/* Paragraph Styling */
```

```
p {
    text-align: center;
    font-style: italic;
    color: gray;
}
```

```
/* Section Styling */
```

```
.form-section {
    background-color: #f2f2f2;
    border: 2px solid black;
    padding: 20px;
    width: 50%;
    margin: 20px auto;
}
```

```
/* Box Model Demonstration */
```

```
.box-model-demo {
    background-color: lightblue;
    border: 2px solid blue;
    padding: 20px; /* Inside spacing */
    margin: 20px; /* Outside spacing */
    text-align: center;
}
```

```
/* Form Styling */
```

```
form {
    margin-top: 20px;
}
```

```
}
```

```
/* Input Styling */
```

```
input, select {  
    width: 100%;  
    padding: 8px;  
    margin-top: 5px;  
    margin-bottom: 15px;  
    border: 1px solid gray;  
    border-radius: 4px;  
}
```

```
/* Submit Button Styling */
```

```
input[type="submit"] {  
    background-color: darkblue;  
    color: white;  
    cursor: pointer;  
}
```

```
input[type="submit"]:hover {  
    background-color: green;  
}
```

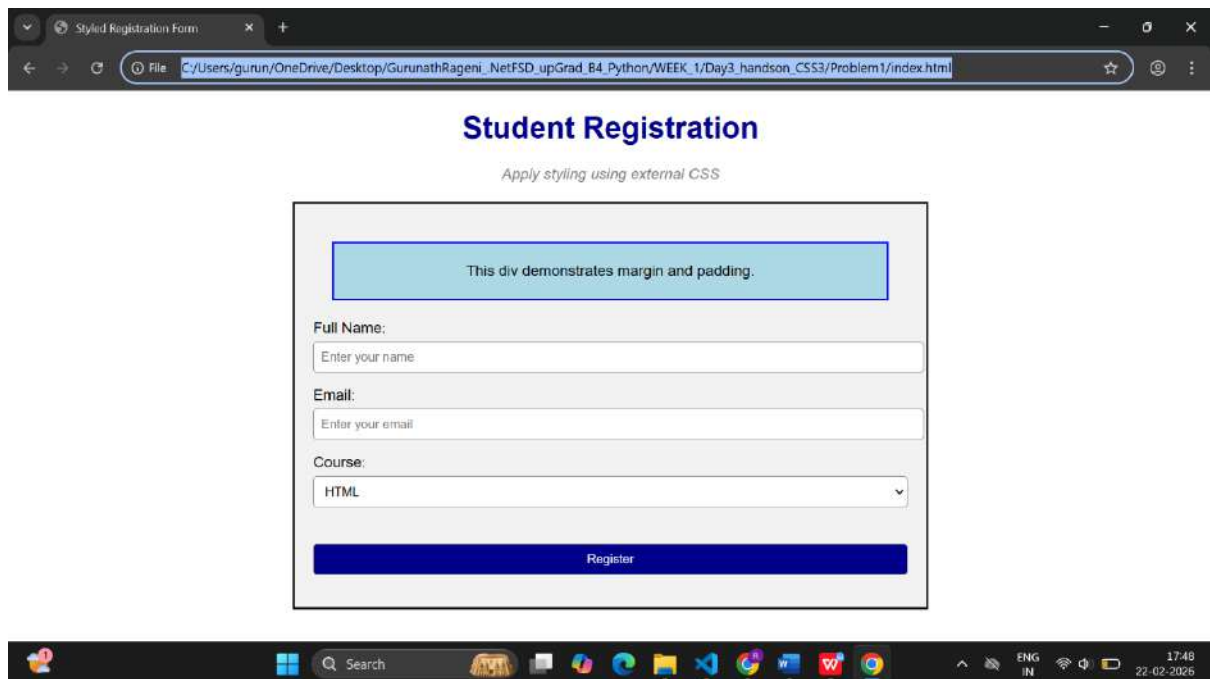
Code Screenshot:

```
index.html X
WEEK_1 > Day3_handson_CSS3 > Problem1 > index.html > html > body > section.form-section > form > label
1  <!DOCTYPE html>
2  <html>
3  <head>
4      <title>Styled Registration Form</title>
5      <link rel="stylesheet" href="style.css">
6  </head>
7  <body>
8
9      <header>
10         <h1>Student Registration</h1>
11         <p>Apply styling using external CSS</p>
12     </header>
13
14     <section class="form-section">
15
16         <div class="box-model-demo">
17             This div demonstrates margin and padding.
18         </div>
19
20         <form>
21             <label>Full Name:</label>
22             <input type="text" placeholder="Enter your name">
23
24             <label>Email:</label>
25             <input type="email" placeholder="Enter your email">
26
27             <label>Course:</label>
28             <select>
29                 <option>HTML</option>
30                 <option>CSS</option>
31                 <option>JavaScript</option>
32             </select>
33
34             <br><br>
35
36             <input type="submit" value="Register">
37         </form>
38
39     </section>
40
41
42 </body>
43 </html>
```

```
# style.css X
WEEK_1 > Day3_handson_CSS3 > Problem1 > # style.css > ...
1  /* Universal Selector */
2  * {
3      font-family: Arial, sans-serif;
4  }
5
6  /* Element Selector */
7  h1 {
8
9      Describes how inline contents of a block are horizontally aligned if the contents do not completely fill the line box.
10     ✓ Widely available across major browsers (Baseline since 2015)
11     Syntax: start | end | left | right | center | justify | match-parent
12 }
13 /* Paragraph Styling */
14 p {
15     text-align: center;
16     font-style: italic;
17     color: gray;
18 }
19
20 /* Section Styling */
21 .form-section {
22     background-color: #f2f2f2;
23     border: 2px solid black;
24     padding: 20px;
25     width: 50%;
26     margin: 20px auto;
27 }
28
29 /* Box Model Demonstration */
30 .box-model-demo {
```

```
30     text-align: center;
31 }
32 /* Form Styling */
33 form {
34     margin-top: 20px;
35 }
36 /* Input Styling */
37 input, select {
38     width: 100%;
39     padding: 8px;
40     margin-top: 5px;
41     margin-bottom: 15px;
42     border: 1px solid gray;
43     border-radius: 4px;
44 }
45 /* Submit Button Styling */
46 input[type="submit"] {
47     background-color: darkblue;
48     color: white;
49     cursor: pointer;
50 }
51 input[type="submit"]:hover {
52     background-color: green;
53 }
```

Output Screenshot:



Explanation:

In this Problem, I created a Student Registration webpage and applied styling using an external CSS file. The HTML structure is organized using semantic elements like `<header>` and `<section>`, and the CSS file is linked using the `<link>` tag. I demonstrated different types of selectors such as universal, element, class, and attribute selectors to style headings, paragraphs, forms, and buttons. The box model concept is clearly shown using a separate div with margin, padding, and border properties. Overall, this project demonstrates the use of external CSS, proper layout styling, and understanding of selectors and the box model to create a visually structured webpage.

Problem 2: Personal Profile Card (Level-1)

Scenario

You are building a simple personal profile card for a portfolio website.

The card will display a user's name, role, short bio, and contact button, styled purely using CSS.

Requirements

Create a profile card layout containing:

- Name (heading)
- Role/title
- Short description paragraph
- Contact button

Apply:

- Text formatting (alignment, decoration)
- Font styles (font-family, size, weight)
- Typography enhancements (line-height, letter-spacing)

Use the CSS Box Model:

- Padding for spacing inside the card
- Border to define the card boundary
- Margin to separate the card from the page edges

Add:

- Background color
- Border radius
- Hover effect on the button

Technical Constraints

- Use only HTML + CSS
- No images or icons
- No frameworks (Bootstrap, Tailwind, etc.)

Code:

```
<!DOCTYPE html>
```

```
<html>
```

```
<head>
```

```
  <title>Personal Profile Card</title>
```

```
  <link rel="stylesheet" href="styles.css">
```

```
</head>
```

```
<body>
```

```
  <div class="card">
```

```
    <h1>Gurunath R</h1>
```

```
    <h3 class="role">Python Developer</h3>
```

```
    <p class="bio">
```

```
      Passionate software developer with strong knowledge in
```

```
      Data Structures, Algorithms, and Full Stack Development.
```


Focused on building scalable and efficient applications.

</p>

<button class="contact-btn">Contact Me</button>

</div>

</body>

</html>

/* Page Styling */

body {

background-color: #f4f6f9;

font-family: Arial, Helvetica, sans-serif;

display: flex;

justify-content: center;

align-items: center;

height: 100vh;

}

/* Profile Card */

.card {

background-color: white;

width: 350px;

/* Box Model */

padding: 25px; /* Inside spacing */

border: 2px solid #ddd; /* Card boundary */

margin: 20px; /* Space from page edges */

border-radius: 12px; /* Rounded corners */

```
    text-align: center;
}
```

```
/* Name Styling */
```

```
.card h1 {
    font-size: 26px;
    font-weight: bold;
    margin-bottom: 5px;
    letter-spacing: 1px;
}
```

```
/* Role Styling */
```

```
.role {
    color: #555;
    font-size: 16px;
    font-weight: 600;
    margin-bottom: 15px;
    text-transform: uppercase;
}
```

```
/* Bio Styling */
```

```
.bio {
    font-size: 14px;
    line-height: 1.6;    /* Typography improvement */
    letter-spacing: 0.5px;
    color: #444;
    margin-bottom: 20px;
}
```

```
/* Button Styling */
```

```
.contact-btn {
```

```
background-color: #1e88e5;
color: white;
padding: 10px 18px;
border: none;
border-radius: 6px;
font-size: 14px;
cursor: pointer;
transition: 0.3s ease;
}
```

```
/* Hover Effect */
```

```
.contact-btn:hover {
    background-color: #961429;
}
```

Code Screenshots:

styles.css

<> index.html X

WEEK_1 > Day3_handson_CSS3 > problem2 > <> index.html >  html

```
1  <!DOCTYPE html>
2  <html>
3  <head>
4  |   <title>Personal Profile Card</title>
5  |   <link rel="stylesheet" href="styles.css">
6  </head>
7  <body>
8  |
9  |   <div class="card">
10 |       <h1>Gurunath R</h1>
11 |       <h3 class="role">Python Developer</h3>
12 |       <p class="bio">
13 |           Passionate software developer with strong knowledge in
14 |           Data Structures, Algorithms, and Full Stack Development.
15 |           Focused on building scalable and efficient applications.
16 |       </p>
17 |
18 |       <button class="contact-btn">Contact Me</button>
19 |   </div>
20 |
21 </body>
22 </html>
23
```

WEEK_1 > Day3_handson_CSS3 > problem2 > # styles.css > .card

```
1  /* Page Styling */
2  body {
3      background-color: #f4f6f9;
4      font-family: Arial, Helvetica, sans-serif;
5      display: flex;
6      justify-content: center;
7      align-items: center;
8      height: 100vh;
9  }
10
11 /* Profile Card */
12 .card {
13     background-color: white;
14     width: 350px;
15
16     /* Box Model */
17     padding: 25px;          /* Inside spacing */
18     border: 2px solid #ddd; /* Card boundary */
19     margin: 20px;           /* Space from page edges */
20
21     border-radius: 12px;    /* Rounded corners */
22
23     text-align: center;
24 }
25
26 /* Name Styling */
27 .card h1 {
28     font-size: 26px;
29     font-weight: bold;
```

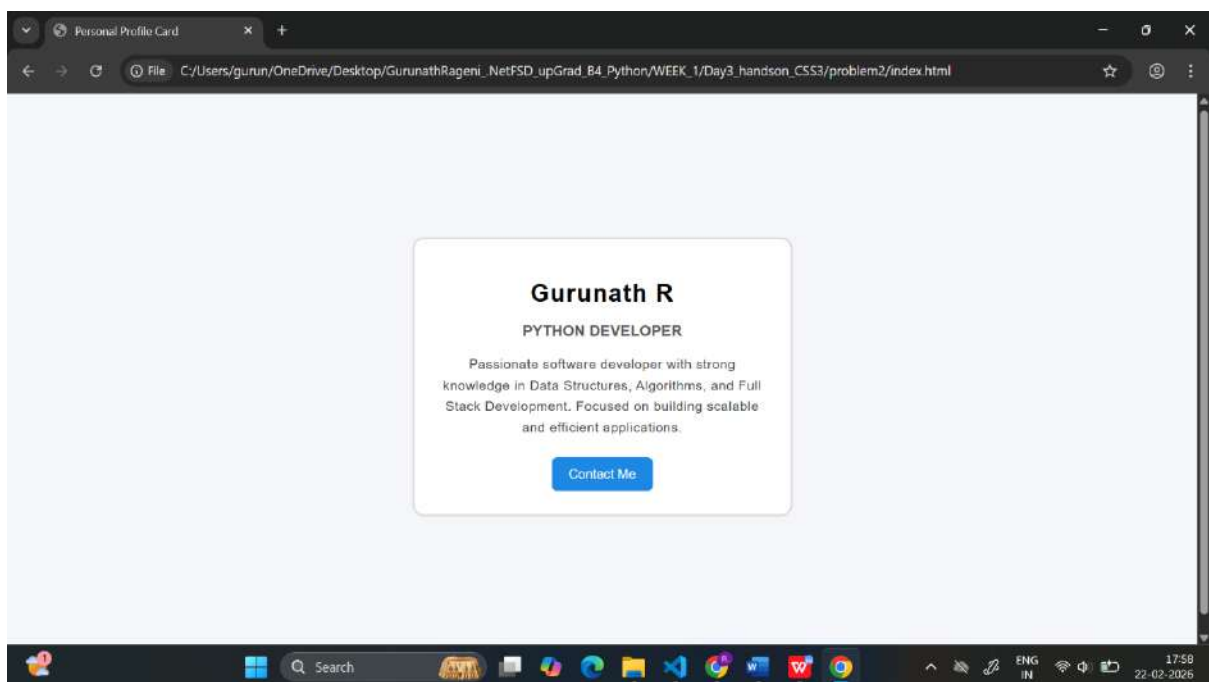
```
29     font-weight: bold;
30     margin-bottom: 5px;
31     letter-spacing: 1px;
32 }
33
34 /* Role Styling */
35 .role {
36     color: █#555;
37     font-size: 16px;
38     font-weight: 600;
39     margin-bottom: 15px;
40     text-transform: uppercase;
41 }
42
43 /* Bio Styling */
44 .bio {
45     font-size: 14px;
46     line-height: 1.6;           /* Typography improvement */
47     letter-spacing: 0.5px;
48     color: █#444;
49     margin-bottom: 20px;
50 }
51
52 /* Button Styling */
53 .contact-btn {
54     background-color: █#1e88e5;
55     color: █white;
56     padding: 10px 18px;
```

```

55     color: white;
56     padding: 10px 18px;
57     border: none;
58     border-radius: 6px;
59     font-size: 14px;
60     cursor: pointer;
61     transition: 0.3s ease;
62 }
63
64 /* Hover Effect */
65 .contact-btn:hover {
66     background-color: #961429;
67 }
68

```

Output Screenshot:



Explanation:

In this Problem, I designed a Personal Profile Card using HTML and styled it with an external CSS file. The layout is centered using Flexbox, ensuring the card appears in the middle of the page both vertically and horizontally. I applied the box model properties such as padding, margin, border, and border-radius to create a clean card design with rounded corners. Typography

enhancements like letter-spacing, line-height, and text transformations improve readability and visual appeal. A styled button with a hover effect and smooth transition adds interactivity, making the profile card look modern and professional.

Problem 3: Product Feature List Section (Level-1)

Scenario

You are designing a feature highlight section for a product landing page that lists key features in a visually appealing way.

Requirements

Create a section with:

- Section heading
- List of 4–5 features

Style the list using:

- Custom fonts
- Text color and background color
- Border and padding for each feature item

Use:

- CSS selectors (element, class, descendant)

Apply hover effects on feature items

Technical Constraints

- Use unordered list ()
- No JavaScript
- No layout frameworks
- Only basic HTML structure

Code:

```
<!DOCTYPE html>
```

```
<html>
```

```
<head>
```

```
  <title>Product Feature Section</title>
```

```
  <link rel="stylesheet" href="styles.css">
```



```
</head>
```

```
<body>
```

```
<section class="features">
```

```
<h2>Product Features</h2>
```

```
<ul>
```

```
<li>Fast Performance</li>
```

```
<li>User-Friendly Interface</li>
```

```
<li>Secure Data Protection</li>
```

```
<li>Cloud Integration</li>
```

```
<li>24/7 Customer Support</li>
```

```
</ul>
```

```
</section>
```

```
</body>
```

```
</html>
```

```
/* Element Selector */
```

```
body {
```

```
  font-family: "Segoe UI", Arial, sans-serif;
```

```
  background-color: #f5f7fa;
```

```
}
```

```
/* Class Selector */
```

```
.features {
```

```
  width: 400px;
```

```
  margin: 40px auto;
```

```
  text-align: center;
```

```
}
```

```
/* Descendant Selector */
```

```
.features h2 {  
    color: #2c3e50;  
    margin-bottom: 20px;  
    letter-spacing: 1px;  
}
```

```
/* List Styling */
```

```
.features ul {  
    list-style-type: none; /* Remove default bullets */  
    padding: 0;  
}
```

```
/* Feature Items */
```

```
.features ul li {  
    background-color: #ffffff;  
    color: #333;  
  
    border: 1px solid #ddd;  
    padding: 10px;  
    margin-bottom: 12px;  
  
    border-radius: 20px;  
    transition: 0.3s ease;  
}
```

```
/* Hover Effect */
```

```
.features ul li:hover {  
    background-color: #2c3e50;  
    color: white;  
    cursor: pointer;  
}
```

Code Screenshot:

```
# styles.css  <> index.html X
WEEK_1 > Day3_handson_CSS3 > Problem3 > <> index.html > html > body
1  <!DOCTYPE html>
2  <html>
3  <head>
4      <title>Product Feature Section</title>
5      <link rel="stylesheet" href="styles.css">
6  </head>
7  <body>
8
9      <section class="features">
10         <h2>Product Features</h2>
11
12         <ul>
13             <li>Fast Performance</li>
14             <li>User-Friendly Interface</li>
15             <li>Secure Data Protection</li>
16             <li>Cloud Integration</li>
17             <li>24/7 Customer Support</li>
18         </ul>
19     </section>
20
21 </body>
22 </html>
23
```

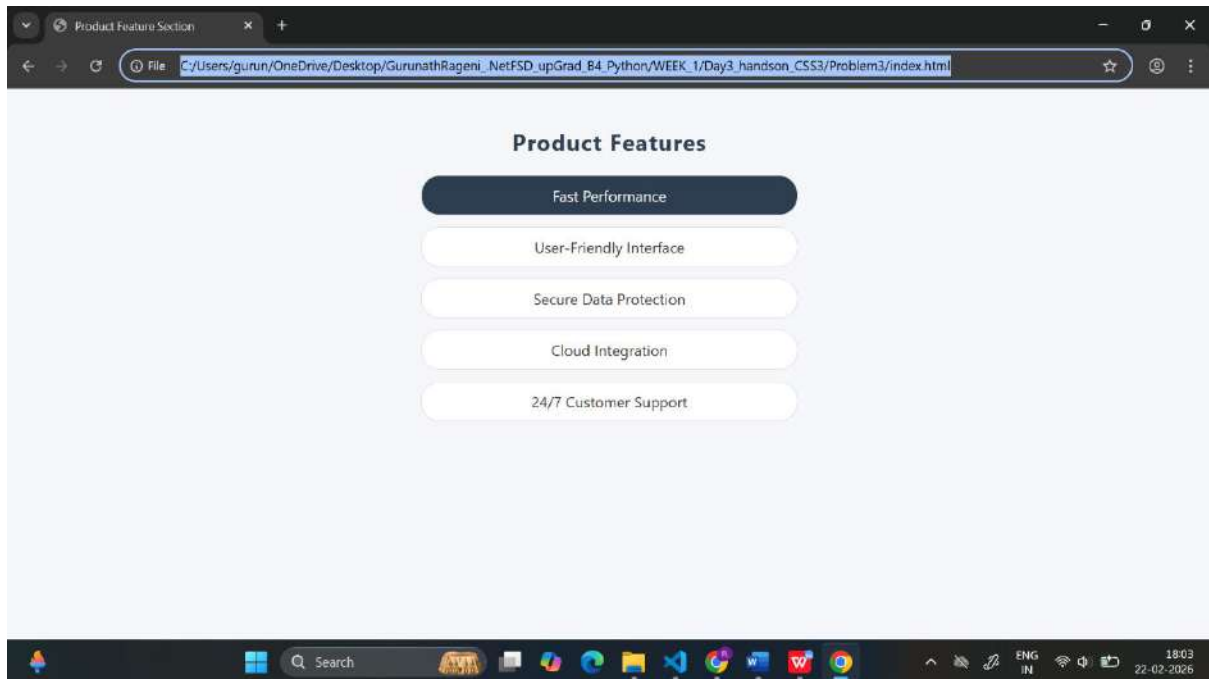
WEEK_1 > Day3_handson_CSS3 > Problem3 > # styles.css > ...

```
1  /* Element Selector */
2  body {
3      font-family: "Segoe UI", Arial, sans-serif;
4      background-color: #f5f7fa;
5  }
6
7  /* Class Selector */
8  .features {
9      width: 400px;
10     margin: 40px auto;
11     text-align: center;
12 }
13
14 /* Descendant Selector */
15 .features h2 {
16     color: #2c3e50;
17     margin-bottom: 20px;
18     letter-spacing: 1px;
19 }
20
21 /* List Styling */
22 .features ul {
23     list-style-type: none; /* Remove default bullets */
24     padding: 0;
25 }
26
```

WEEK_1 > Day3_handson_CSS3 > Problem3 > # styles.css > ...

```
22  .features ul {
23
24  }
25
26
27  /* Feature Items */
28  .features ul li {
29      background-color: #ffffff;
30      color: #333;
31
32      border: 1px solid #ddd;
33      padding: 10px;
34      margin-bottom: 12px;
35
36      border-radius: 20px;
37      transition: 0.3s ease;
38  }
39
40  /* Hover Effect */
41  .features ul li:hover {
42      background-color: #2c3e50;
43      color: white;
44      cursor: pointer;
45  }
46
```

Output Screenshot:



Explanation:

In this Problem I created a Product Feature Section using HTML and styled it with CSS to present features in a clean and modern layout. I applied different types of selectors such as element, class, and descendant selectors to target specific parts of the section. The unordered list is customized by removing default bullets and styling each feature item with padding, borders, and rounded corners. I also implemented hover effects with smooth transitions to enhance interactivity and improve user experience. This project demonstrates the practical use of CSS selectors, layout alignment, and visual enhancements to design an attractive feature section.

Problem 4: Responsive Blog Layout (Level-2)

Scenario

You are creating a blog homepage that displays articles in a modern responsive layout suitable for desktop, tablet, and mobile screens.

Requirements

Create a page layout with:

- Header
- Main content area with multiple blog cards

Use:

- **Flexbox or CSS Grid** for layout
- Box model for spacing

Each blog card should contain:

- Title
- Short description
- Read More button

Apply:

- Background colors
- Borders
- Typography enhancements

Use **media queries** to:

- Display multiple columns on desktop
- Switch to single column on mobile

Technical Constraints

- Only HTML + CSS
- No CSS frameworks
- No fixed widths (use percentages or flexible units)

Code:

```
<!DOCTYPE html>
```

```
<html>
```

```
<head>
```

```
  <title>Responsive Blog Layout</title>
```

```
  <link rel="stylesheet" href="styles.css">
```

```
</head>
```

```
<body>
```

```
  <header>
```

```
    <h1>My Tech Blog</h1>
```

```
    <p>Latest articles on Web Development</p>
```

```
  </header>
```

```
  <main class="blog-container">
```

```
<section class="blog-card">

  <h2>Understanding HTML Basics</h2>

  <p>Learn the building blocks of web development and how HTML structures content on the web.</p>

  <button>Read More</button>

</section>


<article class="blog-card">

  <h2>CSS Flexbox Guide</h2>

  <p>A complete guide to mastering flexible layouts using CSS Flexbox.</p>

  <button>Read More</button>

</article>


<article class="blog-card">

  <h2>Responsive Web Design</h2>

  <p>Build websites that adapt smoothly across mobile, tablet, and desktop devices.</p>

  <button>Read More</button>

</article>


<article class="blog-card">

  <h2>Introduction to JavaScript</h2>

  <p>Understand the basics of JavaScript and how it brings interactivity to websites.</p>

  <button>Read More</button>

</article>


</main>


</body>

</html>
```



```
/* Reset basic spacing */
```

```
*{  
  margin: 0;  
  padding: 0;  
  box-sizing: border-box;  
}
```

```
/* Body Styling */
```

```
body {  
  font-family: Arial, Helvetica, sans-serif;  
  background-color: #f4f6f9;  
  line-height: 1.6;  
}
```

```
/* Header */
```

```
header {  
  text-align: center;  
  padding: 2rem;  
  background-color: #2c3e50;  
  color: white;  
}
```

```
header h1 {  
  margin-bottom: 0.5rem;  
  letter-spacing: 1px;  
}
```

```
/* Blog Container - FLEXBOX */
```

```
.blog-container {  
  display: flex;  
  flex-wrap: wrap;
```

```
    gap: 4rem;
    padding: 2rem;
}
```

```
/* Blog Card */
```

```
.blog-card {
    background-color: white;
    padding: 1.5rem;
    border: 1px solid #ddd;
    border-radius: 8px;

    flex: 1 1 45%; /* Flexible width */
}
```

```
/* Card Typography */
```

```
.blog-card h2 {
    margin-bottom: 0.8rem;
    color: #2c3e50;
}
```

```
.blog-card p {
    margin-bottom: 1rem;
    color: #555;
}
```

```
/* Button */
```

```
.blog-card button {
    padding: 0.5rem 1rem;
    border: none;
    background-color: #3498db;
    color: white;
```

```
border-radius: 5px;

cursor: pointer;

transition: 0.3s ease;

}
```

```
.blog-card button:hover {

background-color: #1f6fb2;

}
```

```
/* ===== */

/* Responsive Design */

/* ===== */
```

```
/* Mobile (default) = 1 column */

.blog-card {

flex: 1 1 100%;

}
```

```
/* Tablet */

@media (min-width: 600px) {

.blog-card {

flex: 1 1 48%;

}

}
```

```
/* Desktop */

@media (min-width: 992px) {

.blog-card {

flex: 1 1 30%;

}

}
```

Code Screenshots:

```
index.html X
WEEK_1 > Day3_handson_CSS3 > Problem4 > index.html > ...
1  <!DOCTYPE html>
2  <html>
3  <head>
4      <title>Responsive Blog Layout</title>
5      <link rel="stylesheet" href="styles.css">
6  </head>
7  <body>
8
9      <header>
10         <h1>My Tech Blog</h1>
11         <p>Latest articles on Web Development</p>
12     </header>
13
14     <main class="blog-container">
15
16         <section class="blog-card">
17             <h2>Understanding HTML Basics</h2>
18             <p>Learn the building blocks of web development and how HTML structures content</p>
19             <button>Read More</button>
20         </section>
21
22         <article class="blog-card">
23             <h2>CSS Flexbox Guide</h2>
24             <p>A complete guide to mastering flexible layouts using CSS Flexbox.</p>
25             <button>Read More</button>
26         </article>
27
28         <article class="blog-card">
29             <h2>Responsive Web Design</h2>
30             <p>Build websites that adapt smoothly across mobile, tablet, and desktop device</p>
31             <button>Read More</button>
32         </article>
33
34         <article class="blog-card">
35             <h2>Introduction to JavaScript</h2>
36             <p>Understand the basics of JavaScript and how it brings interactivity to websi</p>
37             <button>Read More</button>
38         </article>
39
40     </main>
41
42 </body>
43 </html>
44
```

styles.css X

WEEK_1 > Day3_handson_CSS3 > Problem4 > # styles.css > ...

```
1  /* Reset basic spacing */
2  * {
3      margin: 0;
4      padding: 0;
5      box-sizing: border-box;
6  }
7
8  /* Body Styling */
9  body {
10     font-family: Arial, Helvetica, sans-serif;
11     background-color: #f4f6f9;
12     line-height: 1.6;
13 }
14
15 /* Header */
16 header {
17     text-align: center;
18     padding: 2rem;
19     background-color: #2c3e50;
20     color: white;
21 }
22
23 header h1 {
24     margin-bottom: 0.5rem;
25     letter-spacing: 1px;
26 }
27
28 /* Blog Container - FLEXBOX */
29 blog-container {
```

```
29 .blog-container {
30     display: flex;
31     flex-wrap: wrap;
32     gap: 4rem;
33     padding: 2rem;
34 }
35
36 /* Blog Card */
37 .blog-card {
38     background-color: white;
39     padding: 1.5rem;
40     border: 1px solid #ddd;
41     border-radius: 8px;
42
43     flex: 1 1 45%; /* Flexible width */
44 }
45
46 /* Card Typography */
47 .blog-card h2 {
48     margin-bottom: 0.8rem;
49     color: #2c3e50;
50 }
51
52 .blog-card p {
53     margin-bottom: 1rem;
54     color: #555;
55 }
```

```

55     }
56
57     /* Button */
58     .blog-card button {
59         padding: 0.5rem 1rem;
60         border: none;
61         background-color: #3498db;
62         color: white;
63         border-radius: 5px;
64         cursor: pointer;
65         transition: 0.3s ease;
66     }
67
68     .blog-card button:hover {
69         background-color: #1f6fb2;
70     }
71
72     /* ===== */
73     /* Responsive Design */
74     /* ===== */
75
76     /* Mobile (default) = 1 column */
77     .blog-card {
78         flex: 1 1 100%;
79     }
80
81     /* Tablet */
82     @media (min-width: 600px) {
83         .blog-card {

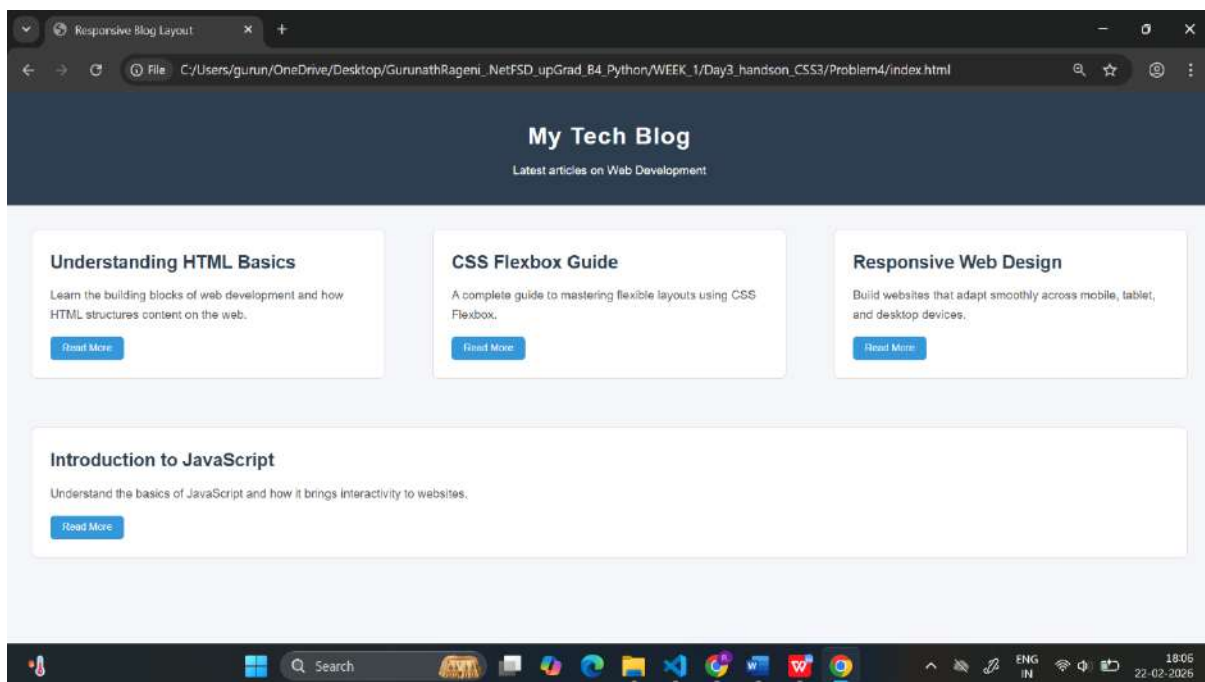
```

```

84         flex: 1 1 48%;
85     }
86 }
87
88 /* Desktop */
89 @media (min-width: 992px) {
90     .blog-card {
91         flex: 1 1 30%;
92     }
93 }
94

```

Output Screenshot:



Explanation:

In this Problem, I designed a Responsive Blog Layout using HTML and CSS with a Flexbox-based structure. The blog posts are organized inside a flexible container that adjusts automatically using flex-wrap and responsive flex properties. I applied a mobile-first approach, where blog cards initially take full width and then adapt to two columns on tablets and three columns on desktops using media queries. Proper spacing, typography, and hover effects enhance readability and user interaction. This project demonstrates the implementation of Flexbox, responsive design principles, and media queries to build a layout that adapts smoothly across different screen sizes.

Problem 5: Responsive Dashboard Layout (Level-2)

Scenario

You are designing a simple admin dashboard layout for a web application.

Requirements

Create a layout with:

- Sidebar navigation
- Main content area
- Info cards inside the main section

Use:

- display properties
- Flexbox or Grid for page structure

Style:

- Sidebar with background color
- Cards with borders, padding, and margins

Apply **media queries** to:

- Convert sidebar into top navigation on mobile
- Stack cards vertically on small screens

Technical Constraints

- HTML + CSS only
- No JavaScript
- No external UI libraries

Code:

```
<!DOCTYPE html>
```

```
<html>
```

```
<head>
```

```
  <title>Responsive Dashboard</title>
```

```
  <link rel="stylesheet" href="styles.css">
```

```
</head>
```

```
<body>
```

```
<div class="dashboard">
```

```
  <!-- Sidebar -->
```

```
  <aside class="sidebar">
```

```
    <h2>Admin Panel</h2>
```

```
    <nav>
```

```
      <a href="#">Dashboard</a>
```

```
      <a href="#">Users</a>
```

```
      <a href="#">Reports</a>
```

```
      <a href="#">Settings</a>
```

```
    </nav>
```

```
  </aside>
```

```
  <!-- Main Content -->
```

```
  <main class="main-content">
```

```
    <h1>Dashboard Overview</h1>
```

```
    <div class="card-container">
```

```
      <div class="card">
```

```
        <h3>Total Users</h3>
```

```
        <p>1,245</p>
```

```
      </div>
```

```
      <div class="card">
```

```
        <h3>Revenue</h3>
```

```
        <p>$8,420</p>
```

```
      </div>
```

```
      <div class="card">
```

```
        <h3>Active Sessions</h3>
```

```
        <p>312</p>
```

```

        </div>

        <div class="card">

            <h3>Pending Orders</h3>

            <p>27</p>

        </div>

    </div>

</main>

</div>

</body>
</html>

/* Reset */
* {
    margin: 0;
    padding: 0;
    box-sizing: border-box;
}

body {
    font-family: Arial, Helvetica, sans-serif;
    background-color: #f4f6f9;
}

/* ===== */
/* Dashboard Layout */
/* ===== */

.dashboard {

```

```
    display: flex;
    min-height: 100vh;
}

/* Sidebar */
.sidebar {
    background-color: #2c3e50;
    color: white;
    padding: 2rem;
    flex: 0 0 220px;
}

.sidebar h2 {
    margin-bottom: 1.5rem;
}

.sidebar nav {
    display: flex;
    flex-direction: column;
}

.sidebar nav a {
    color: white;
    text-decoration: none;
    margin-bottom: 1rem;
    padding: 0.5rem;
    border-radius: 4px;
}

.sidebar nav a:hover {
    background-color: #34495e;
```

```
}
```

```
/* Main Content */
```

```
.main-content {  
  flex: 1;  
  padding: 2rem;  
}
```

```
.main-content h1 {  
  margin-bottom: 2rem;  
}
```

```
/* Cards */
```

```
.card-container {  
  display: flex;  
  flex-wrap: wrap;  
  gap: 1.5rem;  
}
```

```
.card {  
  background-color: white;  
  padding: 1.5rem;  
  border: 1px solid #ddd;  
  border-radius: 8px;  
  
  flex: 1 1 45%;  
}
```

```
.card h3 {  
  margin-bottom: 0.5rem;  
  color: #2c3e50;
```

```
}
```

```
/* ===== */
```

```
/* Responsive Design */
```

```
/* ===== */
```

```
/* Mobile View */
```

```
@media (max-width: 768px) {
```

```
  .dashboard {
```

```
    flex-direction: column;
```

```
  }
```

```
/* Sidebar becomes top navigation */
```

```
.sidebar {
```

```
  flex: none;
```

```
  width: 100%;
```

```
}
```

```
.sidebar nav {
```

```
  flex-direction: row;
```

```
  justify-content: space-around;
```

```
}
```

```
.sidebar nav a {
```

```
  margin-bottom: 0;
```

```
}
```

```
/* Stack cards vertically */
```

```
.card {
```

```
  flex: 1 1 100%;
```

```
}  
  
}
```

Code Screenshots:

```
<> index.html X  
WEEK_1 > Day3_handson_CSS3 > Problem5 > <> index.html > ...  
1  <!DOCTYPE html>  
2  <html>  
3  <head>  
4      <title>Responsive Dashboard</title>  
5      <link rel="stylesheet" href="styles.css">  
6  </head>  
7  <body>  
8  
9      <div class="dashboard">  
10  
11          <!-- Sidebar -->  
12          <aside class="sidebar">  
13              <h2>Admin Panel</h2>  
14              <nav>  
15                  <a href="#">Dashboard</a>  
16                  <a href="#">Users</a>  
17                  <a href="#">Reports</a>  
18                  <a href="#">Settings</a>  
19              </nav>  
20          </aside>  
21  
22          <!-- Main Content -->  
23          <main class="main-content">  
24              <h1>Dashboard Overview</h1>  
25  
26              <div class="card-container">  
27                  <div class="card">  
28                      <h3>Total Users</h3>  
29                      <p>1,245</p>
```

```
29         <p>1,245</p>
30     </div>
31
32     <div class="card">
33         <h3>Revenue</h3>
34         <p>$8,420</p>
35     </div>
36     <div class="card">
37         <h3>Active Sessions</h3>
38         <p>312</p>
39     </div>
40     <div class="card">
41         <h3>Pending Orders</h3>
42         <p>27</p>
43     </div>
44 </div>
45
46 </main>
47
48 </div>
49
50 </body>
51 </html>
```


WEEK_1 > Day3_handson_CSS3 > Problem5 > # styles.css > ...

```
1  /* Reset */
2  * {
3      margin: 0;
4      padding: 0;
5      box-sizing: border-box;
6  }
7
8  body {
9      font-family: Arial, Helvetica, sans-serif;
10     background-color: #f4f6f9;
11 }
12
13 /* ===== */
14 /* Dashboard Layout */
15 /* ===== */
16
17 .dashboard {
18     display: flex;
19     min-height: 100vh;
20 }
21
22 /* Sidebar */
23 .sidebar {
24     background-color: #2c3e50;
25     color: white;
26     padding: 2rem;
27     flex: 0 0 220px;
28 }
29
```

```
29
30 .sidebar h2 {
31     margin-bottom: 1.5rem;
32 }
33
34 .sidebar nav {
35     display: flex;
36     flex-direction: column;
37 }
38
39 .sidebar nav a {
40     color: white;
41     text-decoration: none;
42     margin-bottom: 1rem;
43     padding: 0.5rem;
44     border-radius: 4px;
45 }
46
47 .sidebar nav a:hover {
48     background-color: #34495e;
49 }
50
51 /* Main Content */
52 .main-content {
53     flex: 1;
54     padding: 2rem;
55 }
56
```

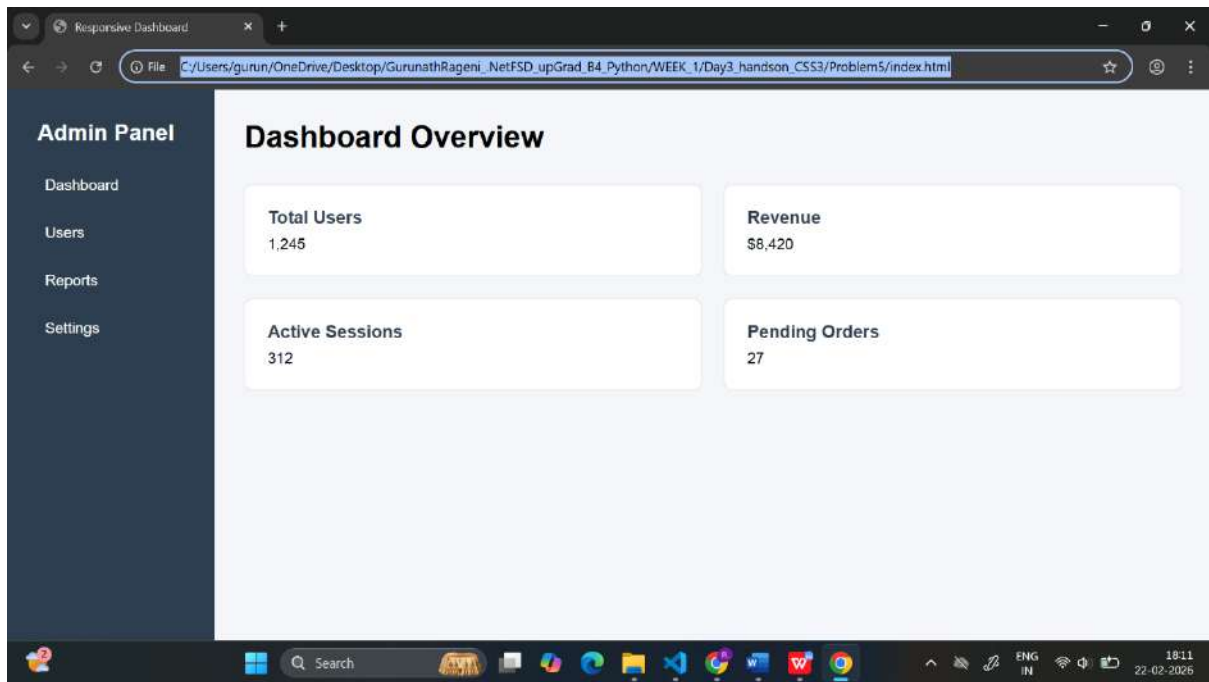
```

57     .main-content h1 {
58         margin-bottom: 2rem;
59     }
60
61     /* Cards */
62     .card-container {
63         display: flex;
64         flex-wrap: wrap;
65         gap: 1.5rem;
66     }
67
68     .card {
69         background-color: #fff;
70         padding: 1.5rem;
71         border: 1px solid #ddd;
72         border-radius: 8px;
73
74         flex: 1 1 45%;
75     }
76
77     .card h3 {
78         margin-bottom: 0.5rem;
79         color: #2c3e50;
80     }
81
82     /* ===== */
83     /* Responsive Design */
84     /* ===== */

```

```
85
86  /* Mobile View */
87  @media (max-width: 768px) {
88
89      .dashboard {
90          flex-direction: column;
91      }
92
93      /* Sidebar becomes top navigation */
94      .sidebar {
95          flex: none;
96          width: 100%;
97      }
98
99      .sidebar nav {
100          flex-direction: row;
101          justify-content: space-around;
102      }
103
104      .sidebar nav a {
105          margin-bottom: 0;
106      }
107
108      /* Stack cards vertically */
109      .card {
110          flex: 1 1 100%;
111      }
112  }
```

Output Screenshot:



Explanation:

In this Problem, I built a Responsive Dashboard layout using HTML and CSS with a clean admin panel structure. The layout is created using Flexbox, where the sidebar and main content are arranged side by side on larger screens. The sidebar contains navigation links, while the main content section displays overview statistics using flexible card components. I used flex-wrap and flexible sizing (flex: 1 1 45%) to ensure the cards adjust properly across different screen sizes. With the help of media queries, the layout becomes mobile-friendly by stacking the sidebar on top and displaying the cards vertically. This project demonstrates responsive layout design, Flexbox usage, and adaptive UI structure for modern web applications.